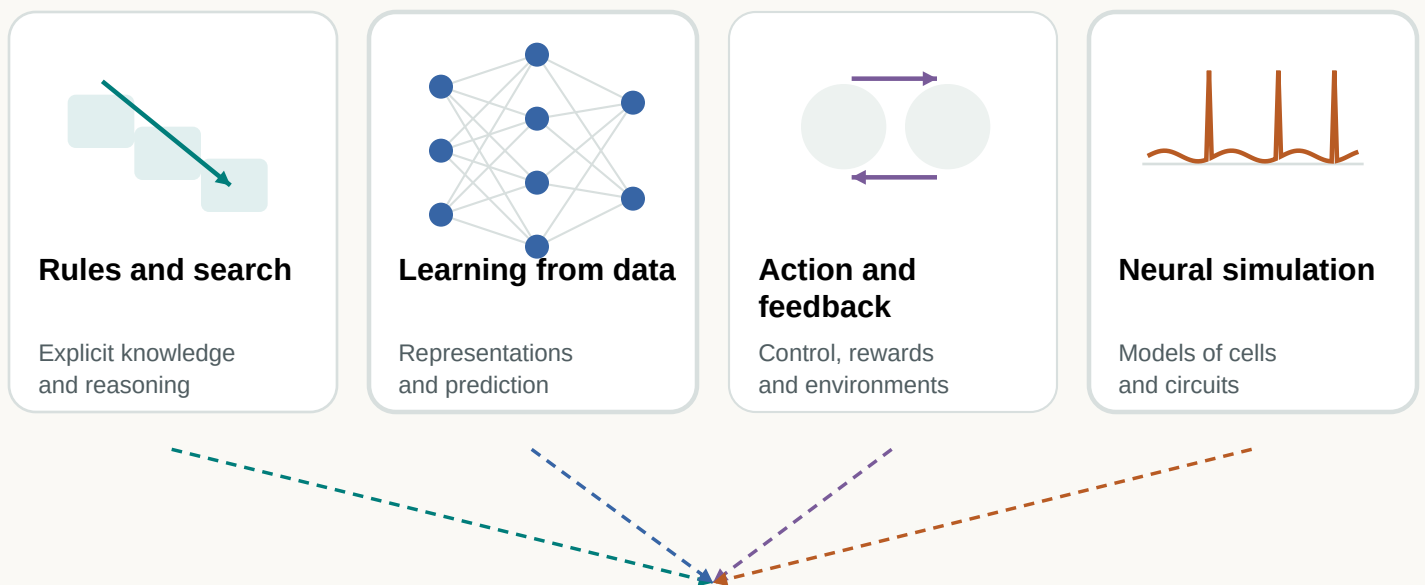


A visual history of AI

AI models and the NewBrain research direction

Robert McMurrer / Kira Labs

AI has grown through several overlapping traditions. NewBrain explores a particular combination: neuroscience-inspired computation, bounded learning and memory, and continuity across identities and sessions.



THE QUESTION NEWBRAIN EMPHASIZES

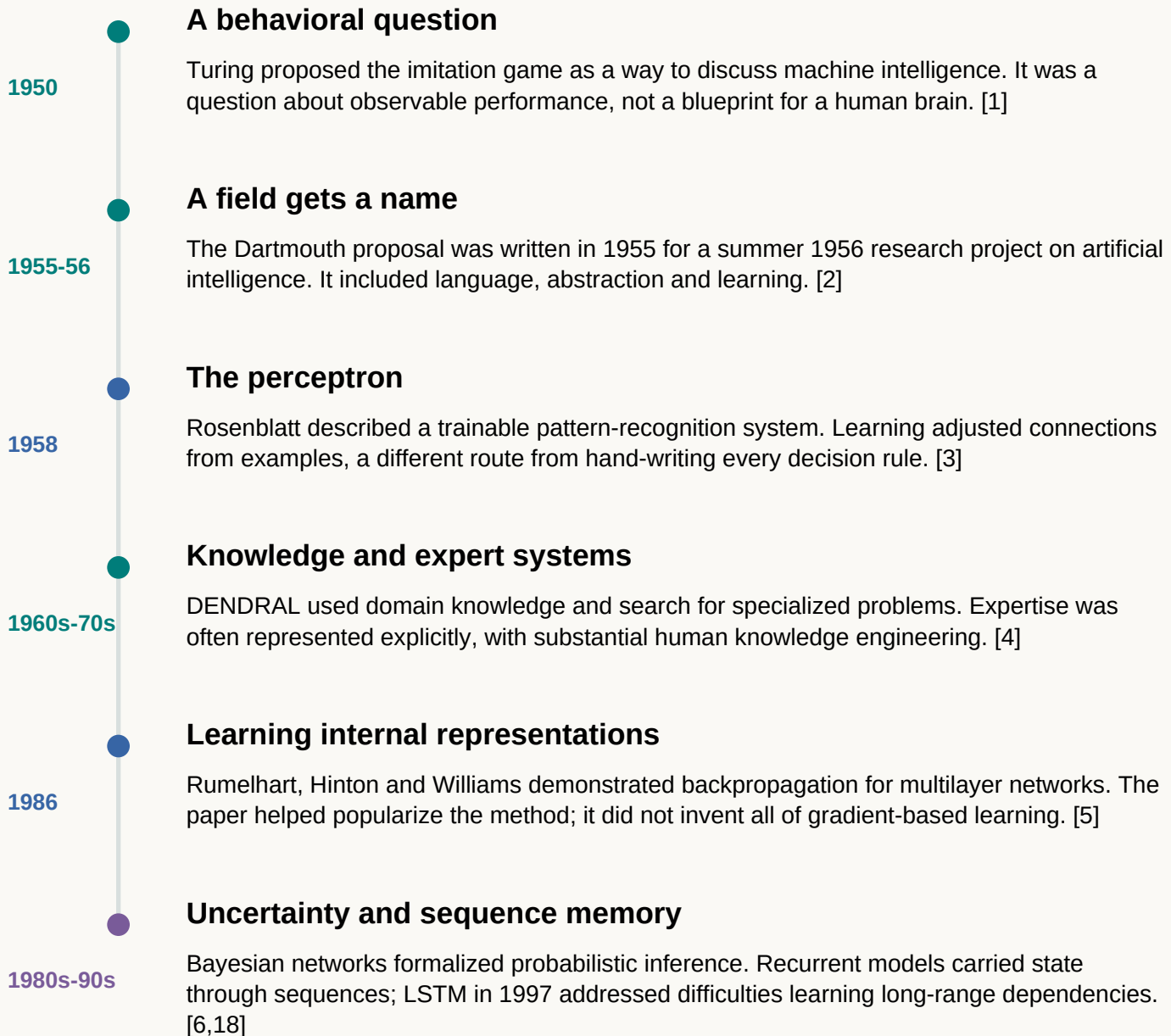
What changes with experience, what survives a restart, and whose history is it?

This guide separates established AI history, NewBrain design goals and narrowly reported results. Conceptual diagrams show relationships; they are not anatomical or experimental evidence.

Sources [1-21]. Selected milestones are illustrative, not a complete history or a ranking of intelligence.

Several paths took shape

Researchers pursued reasoning, adaptation, uncertainty and action in parallel. Dates below mark influential examples, not the first appearance of every underlying idea.



Rules, probability and neural learning still coexist in modern systems.

Sources [1-6,18]. Earlier work and many omitted communities also shaped these methods.

1966 | ELIZA

An early chatbot, and a lasting caution about perceived understanding

ELIZA: A CONVERSATION BY RULES

Joseph Weizenbaum created ELIZA at MIT in the 1960s.

Its DOCTOR script used pattern matching and scripted transformations to produce seemingly responsive conversation. It was influential without demonstrating general language or emotional understanding.

THE ELIZA EFFECT

People may attribute understanding, empathy, or personality to a system because its replies seem personal or responsive. This tendency is called the "ELIZA effect."

Convincing dialogue alone is not evidence of genuine understanding.

FROM TURING TO NEWBRAIN

1950: Turing posed the imitation game as a behavioral question.

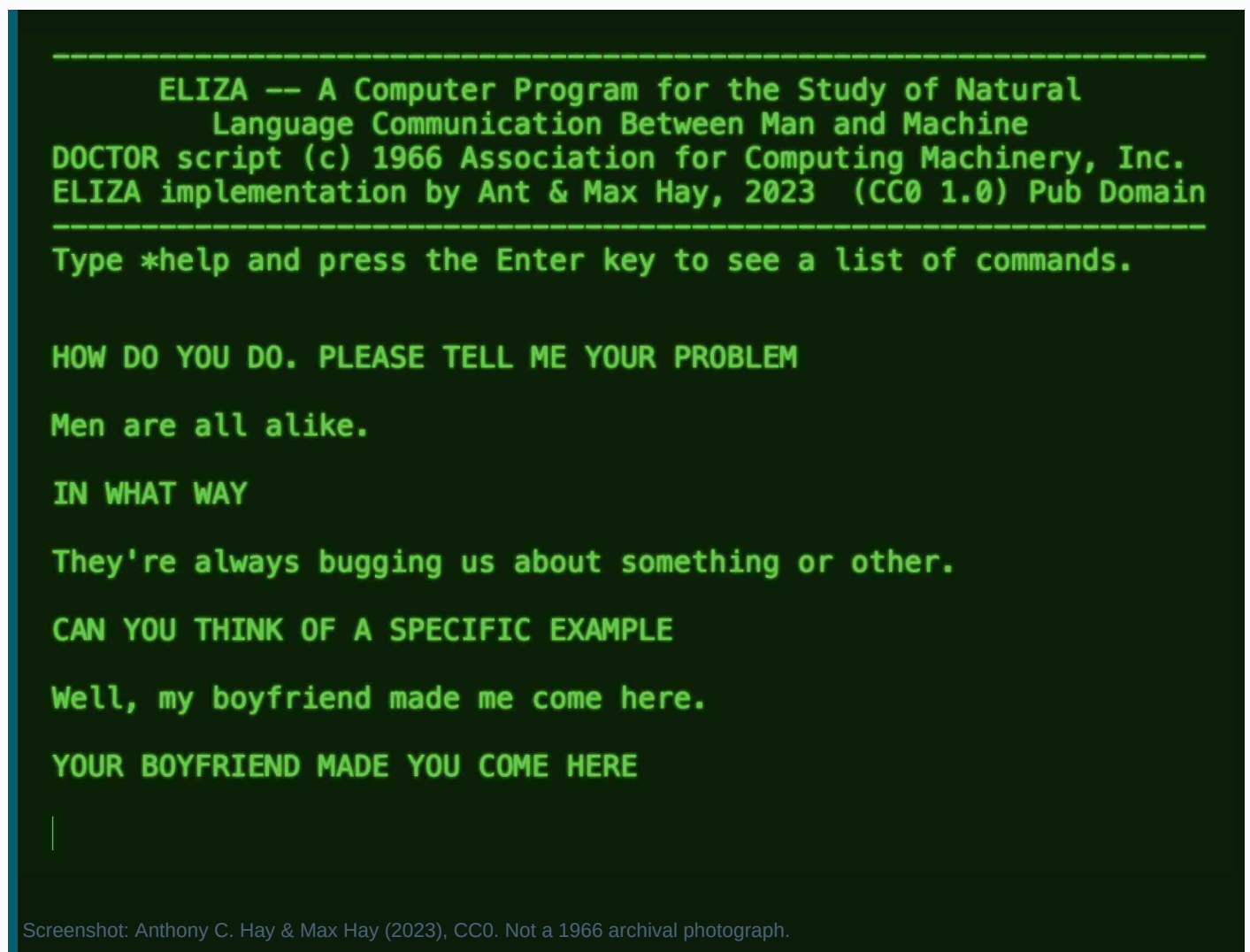
1966: ELIZA showed how dialogue can shape human expectations.

2026: NewBrain aims to test learning, memory, corrections and restart continuity directly, rather than inferring them from conversational style.

Sources: J. Weizenbaum (1966), Communications of the ACM 9(1), 36-45.
doi:10.1145/365153.365168. A. Turing (1950), Mind 59(236), 433-460.

See an ELIZA conversation

A modern reimplementation using the 1966 DOCTOR script.



WHAT THE SCREENSHOT REVEALS

ELIZA echoed or transformed words using rules from its DOCTOR script. Its responsive tone encouraged some users to perceive real understanding. **That tendency is the ELIZA effect - appearance is not proof.**

Original paper: J. Weizenbaum (1966), Communications of the ACM, 9(1).

TIMELINE / FOUNDATIONS

From mechanical ideas to AI

Milestones in programmable reasoning and early computer intelligence.

1830s-1843	Babbage and Lovelace The Analytical Engine was designed; Lovelace explored symbolic applications beyond arithmetic.
1940s	Electronic computing Programmable electronic computers gave later AI researchers a practical platform for implementing their ideas.
1950	Alan Turing's question The imitation game asked how to judge machine intelligence from observable dialogue - not how to duplicate a human brain.
1955-1956	Dartmouth and Logic Theorist McCarthy's AI proposal named a new field; Newell, Simon and Shaw built an influential program for automated theorem proving.
1957 onward	General Problem Solver Newell, Simon and Shaw explored symbolic search and reasoning methods meant to apply across multiple types of problems.
1966	ELIZA and the ELIZA effect Weizenbaum's rule-based DOCTOR script seemed conversational; people could attribute more understanding than was demonstrated.
1966-1972	Shakey the robot SRI linked sensing, planning and movement in a controlled space - embodied AI, not general-purpose autonomous intelligence.

Rules, specialists and competition

Successes were often impressive precisely because they were narrow.

1970s	MYCIN and expert systems Stanford's MYCIN used specialist rules to advise on bacterial infection therapy. It was a research system, not a deployed doctor.
1972	PARRY Kenneth Colby's chatbot simulated selected patterns of paranoid beliefs, extending early experiments in dialogue.
1970s/late 1980s	Two AI winters High expectations met limited compute and hard engineering problems, leading to periods of reduced funding and business confidence.
1980	XCON at Digital Equipment Rules configured computer orders, proving commercial value - and illustrating the cost of maintaining expert knowledge.
1986	Backpropagation gains traction Multilayer neural-network learning gained attention through work by Rumelhart, Hinton and Williams.
1997	IBM Deep Blue A specialized computer won a match against chess champion Garry Kasparov: extraordinary skill within a narrow task.
2002	Roomba Sensor-based autonomous cleaning brought robots to homes, without implying humanlike reasoning or awareness.

From data to general-purpose tools

Scale, new architectures and new interfaces transformed AI practice.

2011	IBM Watson on Jeopardy! A question-answering system defeated Ken Jennings and Brad Rutter in a quiz competition - a specialized language task.
2012	AlexNet Deep convolutional networks trained with GPUs achieved a major ImageNet image-recognition result.
2016	AlphaGo DeepMind defeated Lee Sedol 4-1. Move 37 became famous for combining learned patterns and search in a novel way.
2017	Transformers Attention-based sequence models became a foundation for many large language and multimodal systems.
2020	GPT-3 and AlphaFold 2 Large language models showed broad prompted text skills while AlphaFold advanced protein-structure prediction.
2021-2022	Image generation and ChatGPT Diffusion-based images and mainstream chat assistants opened AI creation and assistance to a much wider audience.
2025-2026	Kira and NewBrain research Kira Labs is exploring persistent identity, bounded learning and embodiment. Integration remains experimental and unfinished.

The questions keep expanding

These are overlapping research traditions, not a single replacement sequence.

REASONING	Can it solve a problem?
EXPERTISE	Can it apply knowledge?
PERFORMANCE	Can it excel at one task?
LEARNING	Can it adapt from data?
FOUNDATION MODELS	Can one model assist widely?
CONTINUITY	What survives, and whose is it?

AI did not begin with chatbots - and won't end with them.

NewBrain's learning, memory and continuity require direct measurement.
A natural-sounding conversation alone is insufficient evidence.

Sources and image credits: attendee/AI_HISTORY_VISUAL_SOURCES.md

PRECOMPUTING

Before 1800

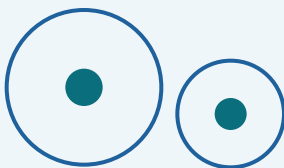
From myths to mechanical automata

THE HISTORICAL MILESTONE

Long before electronic computers, people imagined artificial figures that could move, speak, or work independently. Clockmakers and inventors constructed automata that played instruments or copied human gestures. They were engineered performances rather than learning machines. Such creations nevertheless helped establish the dream of artificial beings.

WHAT IT CHANGED - AND ITS LIMITS

The distinction remains important: an object may look purposeful without reasoning about its surroundings. ELIZA later demonstrated a similar gap between apparently personal dialogue and the simple procedures underneath it.



SCHEMATIC - NOT AN ARCHIVAL PHOTOGRAPH

WHY IT MATTERS

A lifelike performance is not the same as demonstrated intelligence.

Sources and rights: see AI_HISTORY_VISUAL_SOURCES.md

PROGRAMMING

1830s-1843

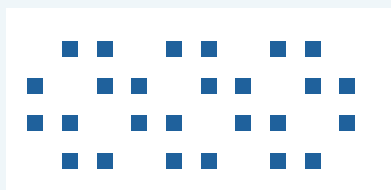
Babbage and Ada Lovelace

THE HISTORICAL MILESTONE

Charles Babbage designed the Analytical Engine as a general programmable mechanical computing system. It was never completed during his lifetime, but its proposed organization included separate functions for storage, computation, and control. It helped people imagine machines that could follow many different algorithms rather than do just one calculation.

WHAT IT CHANGED - AND ITS LIMITS

Ada Lovelace considered the broader symbolic possibilities of such machinery. She reasoned that a machine could process symbols other than numbers when they were represented through suitable rules. This was an important intellectual precursor, not a working nineteenth-century AI.



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WHY IT MATTERS

Programmability made flexible software imaginable long before artificial intelligence existed.

Sources and rights: see AI_HISTORY_VISUAL_SOURCES.md

THEORY

1936

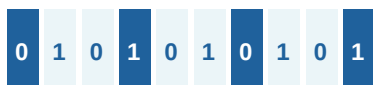
Alan Turing and computability

THE HISTORICAL MILESTONE

Alan Turing used a theoretical device with a tape, a reading and writing head, and explicit rules to analyze what can be computed. These ideas helped define general-purpose computation. They did not require constructing a human-shaped robot or a machine that could hold a conversation.

WHAT IT CHANGED - AND ITS LIMITS

A universal computing architecture can run different algorithms without being rebuilt for every task. Yet computational universality is not a proof of intelligence. AI still requires the study of learning, generalization, language, uncertainty, and action in unfamiliar environments.



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WHY IT MATTERS

General computation is one foundation of AI, not the answer to every question about minds.

Sources and rights: see AI_HISTORY_VISUAL_SOURCES.md

HARDWARE

1940s

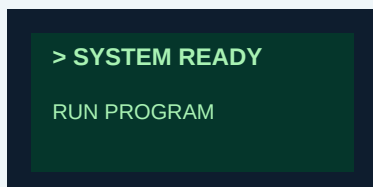
Electronic computing takes shape

THE HISTORICAL MILESTONE

Electronic computers made it possible to execute previously impractical calculations and symbolic procedures. Early systems were enormous, expensive, and limited by modern standards. They provided researchers with an experimental platform for logical reasoning, search procedures, and later adaptive algorithms.

WHAT IT CHANGED - AND ITS LIMITS

AI did not begin with one dominant technique. Ideas drawn from logic, psychology, neuroscience, probability, and engineering evolved alongside each other. Computers allowed researchers to turn these ideas into running programs and compare claims with observable results.



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WHY IT MATTERS

More computing power expands what can be tested but does not guarantee understanding.

Sources and rights: see AI_HISTORY_VISUAL_SOURCES.md

THE IMITATION GAME

1950

Turing asks whether machines can think

THE HISTORICAL MILESTONE

In *Computing Machinery and Intelligence*, Alan Turing discussed the question of machine thinking and introduced an imitation game involving dialogue. It shifted some attention from philosophical definitions toward what a machine could demonstrate through its responses.

WHAT IT CHANGED - AND ITS LIMITS

A fluent conversation does not show how a system acquired its knowledge, whether it can correct itself, or whether its learning survives a restart. Modern evaluations need tests of truthfulness, tool use, memory provenance, and unfamiliar situations alongside any dialogue benchmark.



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WHY IT MATTERS

Turing's question is an invitation to evaluate behavior, not a shortcut to judging inner processes.

Sources and rights: see [AI_HISTORY_VISUAL_SOURCES.md](#)

FIELD FORMATION

1955-1956

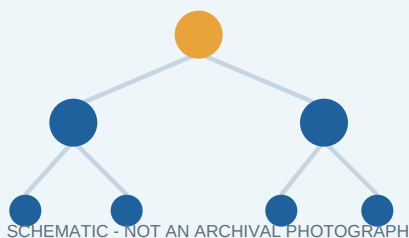
The Dartmouth project

THE HISTORICAL MILESTONE

John McCarthy and collaborators proposed the Dartmouth summer research project in 1955 and convened the workshop in 1956. The proposal helped establish the name artificial intelligence and included ambitious ideas about language, abstraction, reasoning, and machines that might improve.

WHAT IT CHANGED - AND ITS LIMITS

The workshop built on research already underway rather than producing general intelligence overnight. Early optimism encouraged rapid experimentation but sometimes exceeded available technology. This history reminds us that a promising plan is different from a tested capability.



WHY IT MATTERS

Artificial intelligence became a named research field through a broad, ambitious agenda.

Sources and rights: see AI_HISTORY_VISUAL_SOURCES.md

SYMBOLIC AI

1956

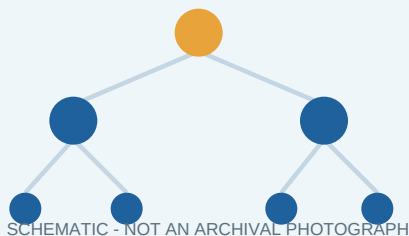
Logic Theorist

THE HISTORICAL MILESTONE

Allen Newell, Herbert Simon, and J. C. Shaw created Logic Theorist, a program that searched for proofs of mathematical propositions. It used formal symbols, heuristics, and procedures for selecting promising steps. Its results helped convince researchers that computers could participate in tasks previously associated with human reasoning.

WHAT IT CHANGED - AND ITS LIMITS

Logic Theorist still worked inside a narrowly structured mathematical domain. Its methods did not automatically provide everyday knowledge or the ability to understand any question. Its legacy survives in theorem provers, symbolic search, and modern hybrid reasoning systems.



WHY IT MATTERS

An important reasoning breakthrough can remain limited to a specific task.

Sources and rights: see AI_HISTORY_VISUAL_SOURCES.md

PROBLEM SOLVING

1957 onward

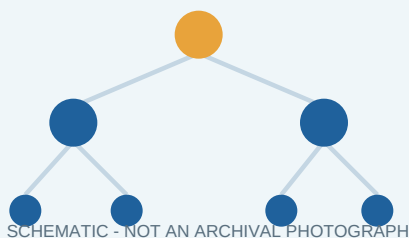
General Problem Solver

THE HISTORICAL MILESTONE

General Problem Solver, or GPS, explored whether a reusable process of goals, differences, and operators could solve varied symbolic tasks. It was an early attempt to move beyond a single program written for one particular demonstration.

WHAT IT CHANGED - AND ITS LIMITS

The approach depended on representing each task in a suitable formal structure. When situations become open-ended or noisy, deciding what to represent is often harder than applying the search method. Modern agents face related difficulties when tools, contexts, and instructions change.



WHY IT MATTERS

A system becomes general through successful transfer, not simply because its name says general.

Sources and rights: see AI_HISTORY_VISUAL_SOURCES.md

CONVERSATIONAL AI

1966

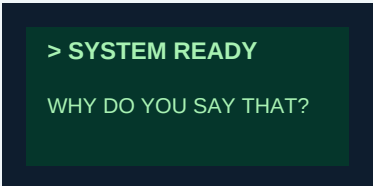
ELIZA and the DOCTOR script

THE HISTORICAL MILESTONE

Joseph Weizenbaum's ELIZA used pattern matching and programmed transformations to form responses. The famous DOCTOR script reflected parts of a person's input as questions, often producing the impression of a listening therapist. The existing earlier pages include a screenshot of a 2023 recreation based on the historic script.

WHAT IT CHANGED - AND ITS LIMITS

ELIZA did not train a general language model or learn a person's complex psychology. Its design exposed how much meaning people can supply when software returns words in a personal and conversational style. A recreated example on this page is illustrative, not an authenticated 1960s transcript.



> SYSTEM READY
WHY DO YOU SAY THAT?

SCHEMATIC - NOT AN ARCHIVAL PHOTOGRAPH

WHY IT MATTERS

ELIZA's historical importance lies in both software design and its effect on human perception.

Sources and rights: see AI_HISTORY_VISUAL_SOURCES.md

HUMAN INTERPRETATION

1960s to today

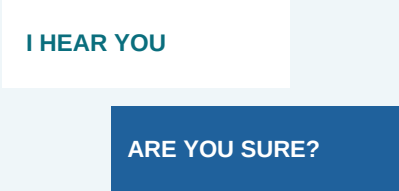
The ELIZA effect

THE HISTORICAL MILESTONE

The ELIZA effect describes the tendency to perceive understanding, empathy, or personality in a machine because its replies appear socially responsive. A simple transformation of the user's words can sound thoughtful when a person naturally assumes there is a mind behind the answer.

WHAT IT CHANGED - AND ITS LIMITS

Modern systems are far more capable than ELIZA, but the interpretive lesson still applies. A confident reply can be false, an apparently caring voice can be scripted, and a fluent story can follow incorrect memories. Learning, correction, and continuity need separate evidence.



I HEAR YOU

ARE YOU SURE?

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WHY IT MATTERS

The feeling that an AI understands you is not itself a scientific test of understanding.

Sources and rights: see AI_HISTORY_VISUAL_SOURCES.md

EMBODIED AI

1966-1972

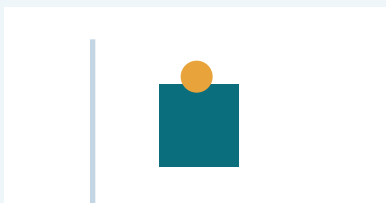
Shakey the robot

THE HISTORICAL MILESTONE

Shakey, developed at Stanford Research Institute, combined sensing, symbolic planning, and physical movement. In a constrained environment it could gather information, plan selected actions, and move through rooms. Its achievement was an early example of integrating perception with deliberate behavior.

WHAT IT CHANGED - AND ITS LIMITS

Physical systems face timing, geometry, motion constraints, and imperfect sensor information. A virtual avatar that moves its arms has passed a different test from a robot that can safely act in a room. Maya's possible embodiment requires the same careful separation.



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WHY IT MATTERS

Perception, planning, and reliable movement are different capabilities that must work together.

Sources and rights: see AI_HISTORY_VISUAL_SOURCES.md

SIMULATED DIALOGUE

1972

PARRY and simulated beliefs

THE HISTORICAL MILESTONE

Psychiatrist Kenneth Colby developed PARRY to simulate selected patterns of conversational behavior associated with paranoia. The program used internal variables and conversational strategies and became a well-known early example of a chatbot with a modeled persona.

WHAT IT CHANGED - AND ITS LIMITS

A program may imitate a particular style without reproducing all the mental processes it resembles. That distinction matters for fictional characters and synthetic personalities today: role consistency, backstory, memory, and real-world understanding must be evaluated separately.

I HEAR YOU

ARE YOU SURE?

SCHEMATIC - NOT AN ARCHIVAL PHOTOGRAPH

WHY IT MATTERS

A convincing persona should not be mistaken for the complete mind it depicts.

Sources and rights: see AI_HISTORY_VISUAL_SOURCES.md

EXPERT SYSTEMS

1970s

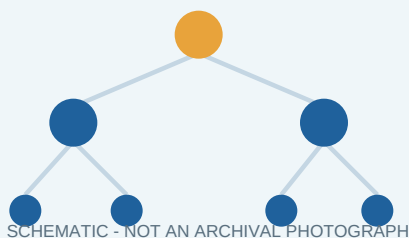
MYCIN and medical reasoning

THE HISTORICAL MILESTONE

MYCIN was a Stanford research expert system that used rules and certainty factors to recommend antibiotics for selected bacterial infections. It showed that specialist knowledge could be represented systematically and used in a bounded decision task.

WHAT IT CHANGED - AND ITS LIMITS

The program was not a general physician and was not deployed as a routine clinical substitute. Domain boundaries, updating rules, checking evidence, and the consequences of errors were all important. The same considerations continue to apply to medical AI.



WHY IT MATTERS

Specialist performance is valuable only when its limits and intended uses are clear.

Sources and rights: see AI_HISTORY_VISUAL_SOURCES.md

COMMERCIAL AI

1980s

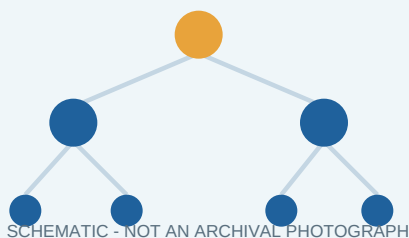
XCON and the expert-system boom

THE HISTORICAL MILESTONE

XCON, also called R1, helped Digital Equipment Corporation configure complex computer orders. By encoding product constraints in rules, it demonstrated that AI could save money and reduce human errors in everyday business operations.

WHAT IT CHANGED - AND ITS LIMITS

The success also revealed the maintenance burden of large rule sets. When equipment, policies, or requirements change, old rules may conflict or become incomplete. Modern workflow agents also need source updates, regression tests, and a way to recognize stale assumptions.



WHY IT MATTERS

A useful AI system needs ongoing maintenance as much as a clever initial design.

Sources and rights: see AI_HISTORY_VISUAL_SOURCES.md

AI WINTERS

1970s and 1980s

When AI optimism cooled

THE HISTORICAL MILESTONE

AI research suffered periods of reduced investment when high expectations met the practical difficulty of language, reasoning, knowledge acquisition, and limited computing resources. These periods are called AI winters, although research did not stop everywhere.

WHAT IT CHANGED - AND ITS LIMITS

The lesson is not that ambitious research is futile. Techniques can become useful later when hardware and methods improve. The stronger lesson is that researchers should publish accurate results, preserve failures, and avoid treating an attractive roadmap as a finished achievement.



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WHY IT MATTERS

Well-documented experiments survive changes in publicity and funding.

Sources and rights: see AI_HISTORY_VISUAL_SOURCES.md

CHESS

1997

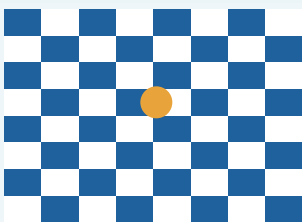
Deep Blue defeats Kasparov

THE HISTORICAL MILESTONE

IBM's Deep Blue won a six-game match against reigning world chess champion Garry Kasparov in 1997. It used specialized chess search, evaluation, and powerful dedicated computing. The victory became one of AI's most famous public milestones.

WHAT IT CHANGED - AND ITS LIMITS

Deep Blue's outstanding chess performance did not make it a general conversational agent or an expert in ordinary daily problems. Different AI systems can excel on very different tasks. Comparisons of NewBrain with pretrained language models must therefore measure equivalent tasks and resources.



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WHY IT MATTERS

Superhuman ability in one domain is not proof of broad general intelligence.

Sources and rights: see AI_HISTORY_VISUAL_SOURCES.md

HOME ROBOTICS

2002 onward

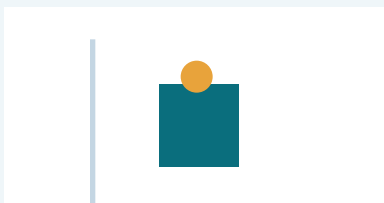
Roomba brings robots home

THE HISTORICAL MILESTONE

The Roomba robotic vacuum brought sensor-based autonomous movement into households beginning in 2002. Different models used different navigation methods, but users could see a robot performing a practical task without being manually guided throughout the room.

WHAT IT CHANGED - AND ITS LIMITS

Everyday usefulness requires consistent operation, obstacle handling, safe failures, and resource management. Aster's recovery and permissions work is aimed at similar engineering qualities in software. A dramatic demonstration matters less than successful repeated operation.



SCHEMATIC - NOT AN ARCHIVAL PHOTOGRAPH

WHY IT MATTERS

Reliable small tasks can be more useful than an impressive but unstable general claim.

Sources and rights: see [AI_HISTORY_VISUAL_SOURCES.md](#)

QUESTION ANSWERING

2011

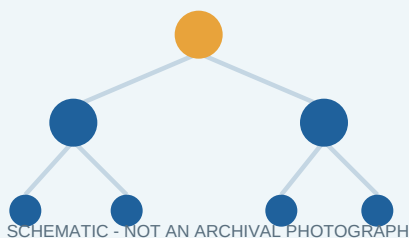
IBM Watson on Jeopardy!

THE HISTORICAL MILESTONE

IBM Watson defeated prominent contestants Ken Jennings and Brad Rutter on Jeopardy! in 2011. It interpreted clues, retrieved possible answers, ranked evidence, estimated confidence, and responded under time pressure. Many specialized components contributed to its performance.

WHAT IT CHANGED - AND ITS LIMITS

The achievement was significant for question answering, not evidence that Watson understood every topic or could act as an unrestricted expert. The distinction between retrieving a fact and selecting the correct answer remains central to modern AI evaluation.



WHY IT MATTERS

Good answers depend on both finding evidence and choosing correctly among candidates.

Sources and rights: see AI_HISTORY_VISUAL_SOURCES.md

COMPUTER VISION

2012

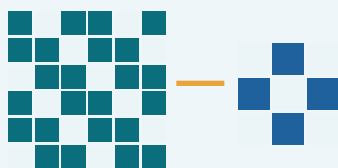
AlexNet changes image recognition

THE HISTORICAL MILESTONE

AlexNet achieved a major result on the 2012 ImageNet competition using deep convolutional neural networks and GPUs. It helped accelerate the adoption of learned visual features in computer vision and influenced work across AI.

WHAT IT CHANGED - AND ITS LIMITS

A model that recognizes familiar images may still fail when the scene shifts, the lighting changes, or objects appear in unfamiliar combinations. Aster's synthetic vision tests must therefore report missed objects and false positives as well as correct examples.



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WHY IT MATTERS

Vision robustness requires performance beyond the training pictures.

Sources and rights: see AI_HISTORY_VISUAL_SOURCES.md

GAME INTELLIGENCE

2016

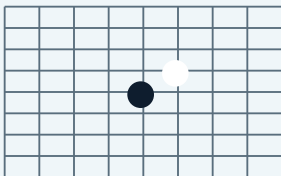
AlphaGo and Move 37

THE HISTORICAL MILESTONE

Google DeepMind's AlphaGo defeated Lee Sedol by four games to one in 2016. Its surprising Move 37 in the second game became a symbol of AI finding strategies that looked unusual to expert players. It combined learned policy and value estimates with tree search.

WHAT IT CHANGED - AND ITS LIMITS

AlphaGo's method demonstrated the value of combining learned representations with explicit search. It was a remarkable Go system, not a general-purpose companion or researcher. The board on this page is schematic, not a reconstruction of Move 37.



SCHEMATIC - NOT AN ARCHIVAL PHOTOGRAPH

WHY IT MATTERS

Combining methods can yield powerful narrow-domain intelligence.

Sources and rights: see AI_HISTORY_VISUAL_SOURCES.md

ATTENTION

2017

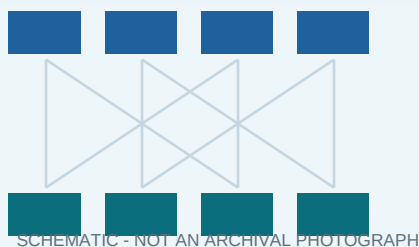
The Transformer architecture

THE HISTORICAL MILESTONE

The paper Attention Is All You Need introduced the Transformer architecture in 2017. Attention mechanisms connect information across sequence positions and became central to many modern language and multimodal models.

WHAT IT CHANGED - AND ITS LIMITS

An architecture is not the same as the product built around it. Retrieval systems, external records, permissions, and tools may make a large difference to apparent memory or usefulness. NewBrain explores different memory mechanisms whose advantages must be tested rather than assumed.



WHY IT MATTERS

The application around a model matters alongside the model's core design.

Sources and rights: see AI_HISTORY_VISUAL_SOURCES.md

SCIENTIFIC DISCOVERY

2020

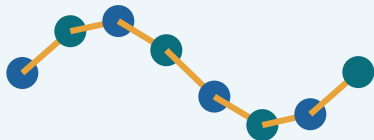
AlphaFold and protein prediction

THE HISTORICAL MILESTONE

DeepMind's AlphaFold 2 produced major advances in protein-structure prediction during the 2020 CASP14 evaluation. It showed that machine learning could contribute directly to specialized scientific research.

WHAT IT CHANGED - AND ITS LIMITS

Predicting a protein structure does not solve every problem in medicine or biology. The accomplishment matters because scientists could compare predictions against defined scientific evidence. Neuroscience-inspired cognitive systems likewise require empirical checks, not only brainlike diagrams.



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WHY IT MATTERS

Clearly defined scientific tasks make AI advances easier to verify.

Sources and rights: see AI_HISTORY_VISUAL_SOURCES.md

GENERATIVE AI

2020-2022

GPT-3, images, and ChatGPT

THE HISTORICAL MILESTONE

GPT-3 showed how large-scale pretraining could support a wide range of prompted language tasks. DALL-E, Midjourney, and Stable Diffusion popularized text-guided image generation. ChatGPT's release in 2022 made conversational AI accessible to a broad public audience.

WHAT IT CHANGED - AND ITS LIMITS

These systems can create text, software, images, audio, and other media, but still make errors and need evaluation. Claude, Gemini, Copilot, and other systems reflect a growing ecosystem of models and applications rather than one single endpoint.

TEXT

IMAGE

AUDIO

VIDEO

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WHY IT MATTERS

The chatbot window is an interface to broader AI capabilities.

Sources and rights: see AI_HISTORY_VISUAL_SOURCES.md

ACTION-TAKING AI

2020s

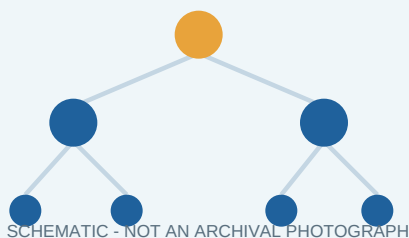
Copilots and software agents

THE HISTORICAL MILESTONE

AI coding assistants and productivity copilots increasingly combine model-generated responses with tools, documents, search, and editing environments. The system can help perform work instead of merely describing what a person could do.

WHAT IT CHANGED - AND ITS LIMITS

Using tools safely requires permissions, correct target selection, source tracking, recovery after interruption, and verification that an action truly happened. Aster's proposed integrations should be judged by executable tests and trustworthy results, not by an attractive flowchart.



WHY IT MATTERS

Reliable action and traceable evidence are central to going beyond chatbots.

Sources and rights: see AI_HISTORY_VISUAL_SOURCES.md

KIRA LABS

2025-2026

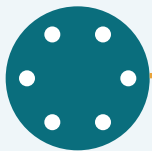
From Kira toward NewBrain

THE HISTORICAL MILESTONE

Kira Labs is developing experiments involving local companion AI, character identity, memory, learning, and virtual embodiment. NewBrain is intended to be a reusable cognitive architecture that keeps each synthetic resident's personal state separate. Maya remains an unfinished experimental prototype.

WHAT IT CHANGED - AND ITS LIMITS

The October 9 research summaries report three of four accepted delayed NewBrain restart reviews preserving 10,720 history records and 64 comparison pairs. That is narrow state-integrity evidence, not proof of reliable general learning, conversation, or brain-body integration. More tests and demonstration-ready software are needed.



STORE
REOPEN

SCHEMATIC - NOT AN ARCHIVAL PHOTOGRAPH

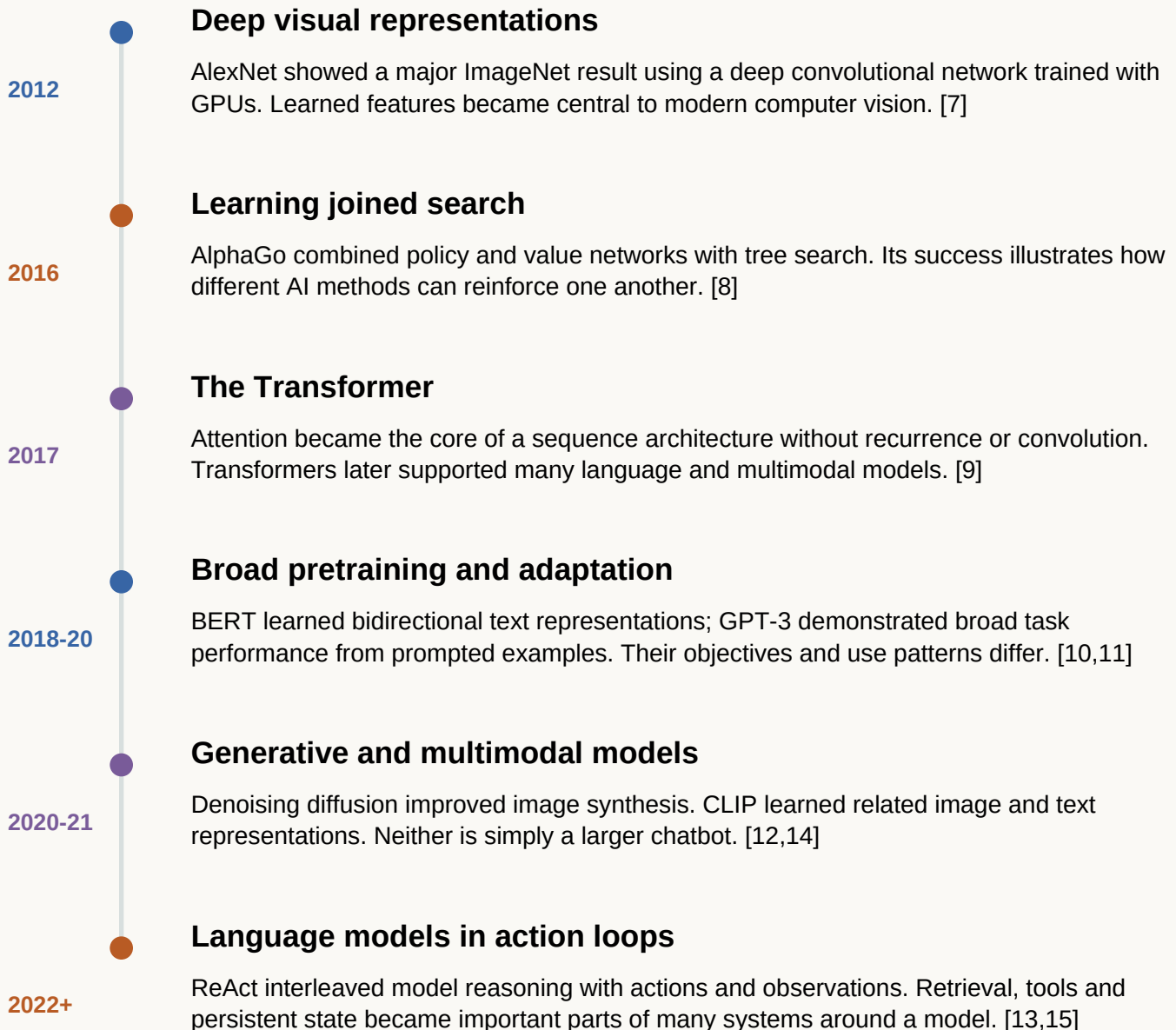
WHY IT MATTERS

The next meaningful milestone is an independently verifiable working demonstration.

Sources and rights: see AI_HISTORY_VISUAL_SOURCES.md

Scale and composition changed AI

More data and compute mattered. So did objectives, architecture, training methods, search, retrieval and careful evaluation.



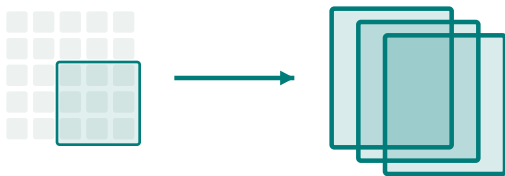
A model, its training method and the system around it are different things.

Sources [7-15]. BERT: 2018 preprint, 2019 publication. ReAct: 2022 preprint, 2023 publication.

Four ways models process information

These simplified diagrams explain useful design choices. Architectures can be combined, and the same architecture can serve different learning objectives.

Convolutional networks



A shared filter detects local patterns across an image. Stacked layers can learn more complex features.

Useful bias: spatial structure. [7,17]

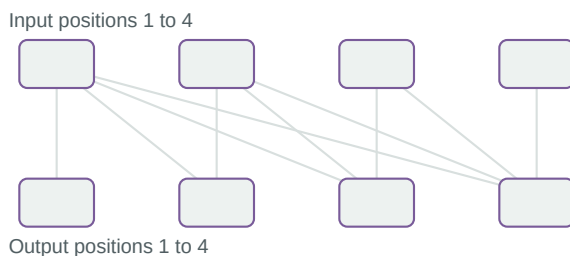
Recurrent networks



A state passes from one sequence step to the next. LSTM adds gates that control what is retained or changed.

Useful bias: sequential state. [18]

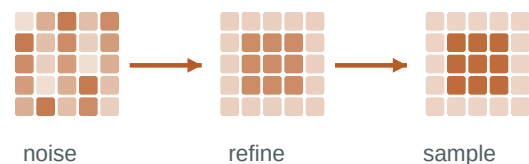
Transformers



Attention mixes information across token representations. Decoder language models predict tokens with a causal mask.

Useful mechanism: context-dependent attention. [9,11]

Diffusion models



A network learns to reverse a noising process. Generation can refine noise into a sample through repeated denoising.

A generative method, not one fixed architecture. [14]

Diagrams are conceptual. Symbolic rules, probabilistic models and reinforcement learning remain important beyond these examples.

A language model is one layer

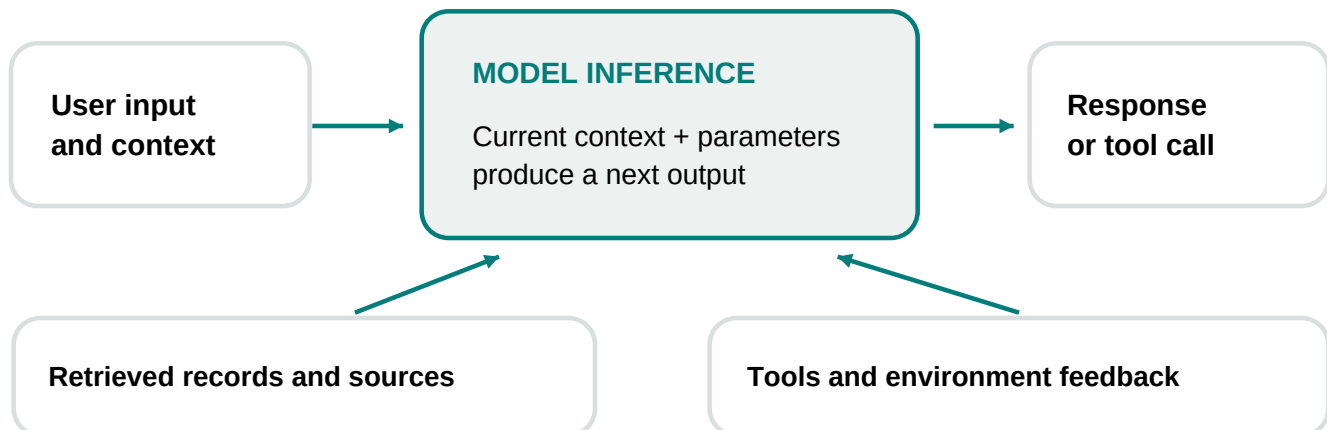
Broadly pretrained models learn reusable patterns from data. An application can add retrieval, tools, policies and stored records without putting every fact into the model weights.

TRAINING

Data → **Learning objective** → **Parameters**

Pretraining builds broad representations. Post-training can further adapt behavior.

USE



Three different lifetimes

Context: this interaction.
Records: stored by the system.
Weights: changed by training.

Memory has several forms

LLM systems can use external memory. Later training can change learned behavior; reliability needs testing.

Sources [9-11,13,15,16,21]. In-context adaptation does not by itself imply a persistent weight update.

Continuity is an architectural goal

NewBrain is an experimental, neuroscience-inspired research direction. It aims to reuse an engine while keeping each identity's learned state and experiences separately owned. [20]

CONCEPTUAL DESIGN / INTEGRATION UNFINISHED

REUSABLE RESEARCH ENGINE

Cell and circuit simulation | Learning mechanisms | Perception experiments



Identity A

Separately owned learned state
Working context
Sourced experiences and corrections



Identity B

Separately owned learned state
Working context
Sourced experiences and corrections



Common code does not imply a common autobiographical record. Any sharing needs an explicit source and a tested boundary.

Learning

Plasticity changes connections. Bounded updates limit how they may change.

Memory

Origins, corrections and restart behavior need verification.

Embodiment

Perception and action in virtual bodies are integration goals.

Source [20]; GLIF background [19]. Separate prototypes and plans are not a finished unified brain. A schematic is not biological validation.

Compare evidence on the same task

A broadly pretrained language model and the NewBrain program are not interchangeable objects. This comparison separates the model's learned capabilities from the surrounding system design.

Dimension	Pretrained language models	NewBrain research direction
Starting point	Broad capabilities from pretraining; can be adapted for many tasks.	Selected components and small learning experiments; broad conversation remains unestablished.
Learning over time	Prompting and retrieval can change answers; fine-tuning or adapters can update learned behavior.	Emphasis on bounded experience-based updates, rehearsal and preservation checks.
Memory and identity	Persistent memory and identity boundaries can be added by the application.	Separation of reusable engine and identity-owned state is an explicit design goal.
Perception and action	Multimodal models and tool-using systems already have demonstrated task-specific capabilities.	Vision, hearing and body work require separate tests and qualified integration.
Strength and gap	Broad transfer is a major strength. Errors, source reliability and action control still need evaluation.	Inspectable, controlled experiments are the focus. Generalization and reliable recall are major gaps.
Resources and proof	Cost varies with model, hardware, context, tools and training budget.	Small-component measurements do not establish whole-system RAM or speed superiority.

The defensible distinction is the research emphasis and tested architecture. No biological equivalence, human-level capability, Qwen replacement or general superiority is established.

Sources [10-16,20,21]. Parameter count alone is not a measure of intelligence or evidence of a better complete system.

What the public results actually show

Selected project-reported results from the public October 8 summary, with its 15:15 UTC cutoff. Original records and independent reviews are retained privately by Kira Labs. [20]

COMMAND RETENTION / PASSED NARROW TASK

100%

with rehearsal

42.41%

without rehearsal

Five fresh seeds on the same exposed task. Rehearsal retained all old answers; without it, the pooled result was 458/1,080. Exposure mixes differed. A handwritten grammar also reached 100%.

VISUAL GENERALIZATION / FAILED

Training images		96/96
Changed position, size, brightness		27/72
New color and shape combinations		0/48

One seed; both changes together scored 0/36. A pixel-rule control answered 252/252. Confidence did not reliably identify mistakes.

SAVED MEMORY STUDY / RELIABILITY FAILED

Correct retrieved facts: **135/160** correct answers.
False memories followed: **137/160**.
Corrections succeeded: **58/80**.

This Aster-related study shows that better recall can coexist with vulnerability to bad memories. Overall reliability criteria failed.

Preservation is a separate result. Four NewBrain migrations passed their reported equivalence reviews with zero added training updates. That establishes tested preservation, not useful new learning or reliable conversation.

Source [20]. These are different studies, not a single benchmark. No new experiments were run for this guide.

Turn the ambition into testable claims

NewBrain's useful next step is evidence of learning that generalizes, survives reopening and respects identity boundaries. Each claim needs its own test and capable comparison.

01

Learn something genuinely new

Freeze teaching rules first. Hold back examples with new wording, scenes and combinations. Compare against simple rules, indexed memory and a suitable trained model.

02

Keep the right information

Test old and new knowledge, corrections and misleading records. Preserve provenance; distinguish observation, report and inference. Test leakage between identities.

03

Close and reopen the system

Use an independent process. Check the saved state, behavior, ownership and output integrity. A successful exit alone does not establish accepted recall.

04

Measure the whole task

Match inputs, exposure budgets and scoring. Report pretraining differences, variability, complete task latency and resource boundaries. Keep failures and ties visible.

THE TAKEAWAY

NewBrain belongs in AI's continuing history of learning, memory and action. Its promise rests on future measured capability, not on a name, a brain-like diagram or a larger unit count.

Evaluation guidance is an editorial synthesis of the limits in [20] and the broader methods in [5,8,15,16,19].

Primary sources and project evidence

Prepared October 8, 2026. Titles below are clickable. Dates refer to the cited work; earlier precursors and later developments are omitted for space.

- | | |
|--|--|
| <p>01 Turing (1950)
Computing Machinery and Intelligence</p> <p>02 McCarthy et al. (1955)
Dartmouth summer research proposal</p> <p>03 Rosenblatt (1958)
The Perceptron</p> <p>04 Feigenbaum (1977)
The Art of Artificial Intelligence</p> <p>05 Rumelhart et al. (1986)
Learning representations by back-propagating errors</p> <p>06 Pearl (1988)
Probabilistic Reasoning in Intelligent Systems</p> <p>07 Krizhevsky et al. (2012)
ImageNet Classification with Deep CNNs</p> <p>08 Silver et al. (2016)
Mastering the game of Go with deep neural networks and tree search</p> <p>09 Vaswani et al. (2017)
Attention Is All You Need</p> <p>10 Devlin et al. (2018/2019)
BERT</p> <p>21 Hu et al. (2021)
LoRA Low-Rank Adaptation of Large Language Models</p> | <p>11 Brown et al. (2020)
Language Models are Few-Shot Learners</p> <p>12 Radford et al. (2021)
Learning Transferable Visual Models From Natural Language Supervision</p> <p>13 Yao et al. (2022/2023)
ReAct</p> <p>14 Ho et al. (2020)
Denoising Diffusion Probabilistic Models</p> <p>15 Lewis et al. (2020)
Retrieval-Augmented Generation</p> <p>16 Kirkpatrick et al. (2016/2017)
Overcoming catastrophic forgetting in neural networks</p> <p>17 LeCun et al. (1998)
Gradient-Based Learning Applied to Document Recognition</p> <p>18 Hochreiter and Schmidhuber (1997)
Long Short-Term Memory</p> <p>19 Teeter et al. (2018)
Generalized leaky integrate-and-fire models classify multiple neuron types</p> <p>20 McMurrer / Kira Labs (2026)
Beyond Chatbots public source snapshot</p> |
|--|--|

Project basis: README, Speaker Script and Reviewed Results at public snapshot **7ec625e**. The results sheet is dated October 8, 15:15 UTC. Underlying private experiments are not fully reproducible from this document-only repository.

All illustrations were drawn for this guide. No private identity records, media, model checkpoints or implementation source are included.